

REVANTAGE CREDENTIALING APP

Developer Functional Flow Document

Purpose: explain how the credentialing app is expected to be used from daily operations, workflow automation, case tracking, email handling, delay ownership, approval and billing readiness perspectives.

Important design direction

This application should be built as a credentialing operations platform, not as a static provider database, not as only document storage, and not as a marketing analytics tool. The system should help credentialing users move each provider/payer application case from intake to billing readiness while preserving communication, documents, tasks, audit trail and accountability.

1. Document Scope and Assumptions

- This flow document explains how the Revantage Credentialing App should be used by credentialing executives, managers, administrators and billing/read-only users.
- The app should support internal credentialing operations first. External provider/practice access can be added later if approved.
- The core operational unit is not only a Provider. The core working unit is a Provider + Practice/Location + Payer Application Case.
- Every meaningful action should update the case timeline and audit trail: status changes, emails, documents, tasks, follow-ups, delay owner changes and approvals.
- Workflow configuration should not be hard-coded. Statuses, document types, SLA days, delay rules, payer templates and email templates should be configurable from Admin Settings.

2. Core Product Concept for Developer

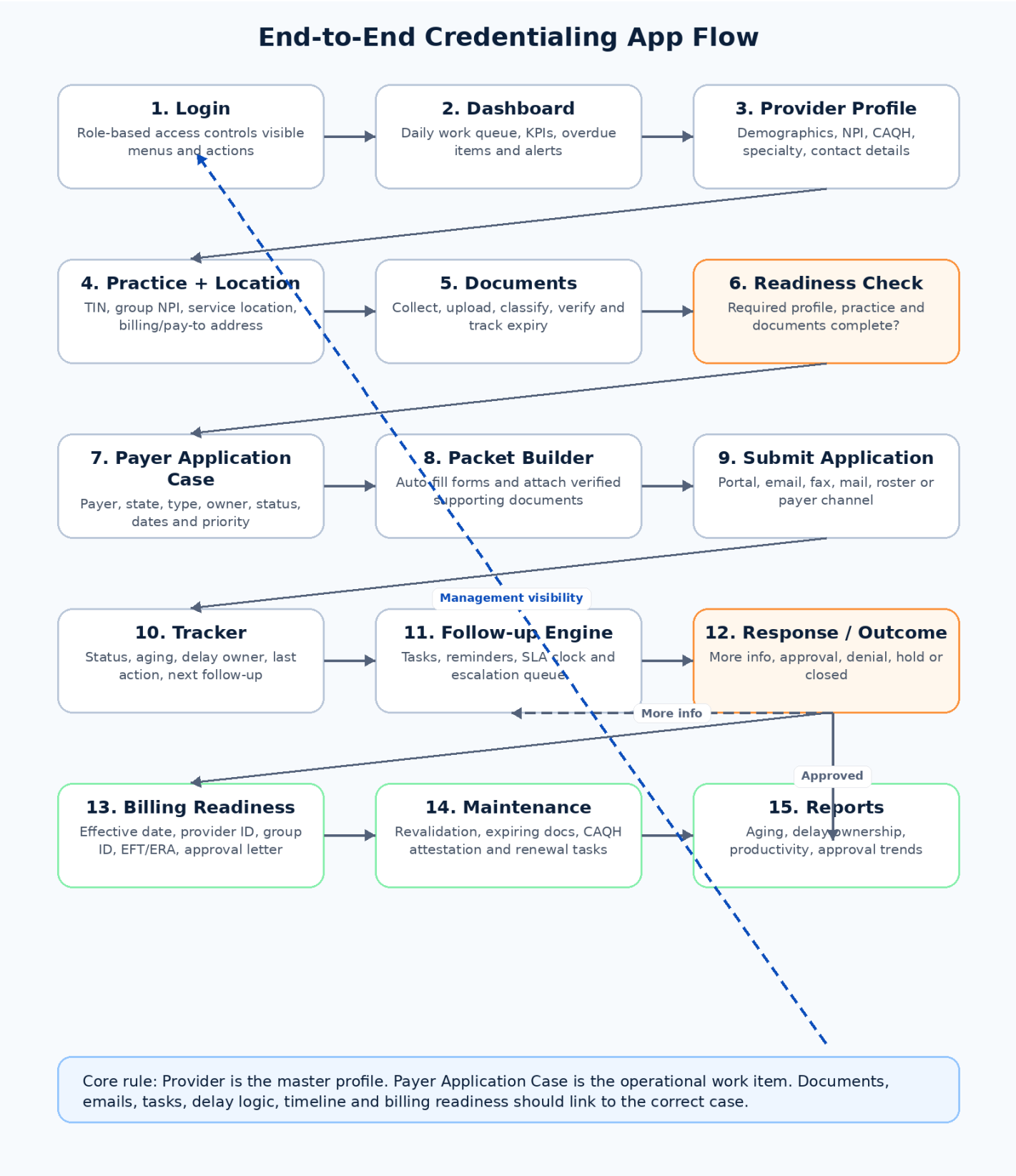
Relationship model

Provider is the master profile. Practice and location define where/how the provider works. Payer Application Case is the operational case for credentialing/enrollment. Documents, emails, tasks, delay logic, timeline and billing readiness should attach to the correct payer case where possible, and to the provider when general.

Object	What it means	Why it matters
Provider	The individual clinician record: demographics, NPI, CAQH, taxonomy, specialty, contact details, IDs.	This is the central master profile used across all payer applications.
Practice	Legal entity/group details: TIN, group NPI, DBA, contacts, addresses.	Credentialing is usually performed under a group/practice relationship.
Location	Service location, billing address, pay-to address, state and contact mapping.	Many payers require location-specific enrollment or validation.
Payer Application Case	One payer/state/application-type workflow for one provider and practice/location combination.	This is the main work item tracked by credentialing users.
Document	Any supporting item such as license, DEA, malpractice, W-9, CV, EFT/ERA, approval letter.	Documents drive readiness, submission and future expiry alerts.
Email Thread	Inbound/outbound credentialing email linked to provider and payer case.	Communication should become part of the case history and not remain separate in Outlook/Gmail.
Task	Manual or system-generated action item with owner, due date and priority.	Prevents missed follow-ups, document requests and escalations.
Delay Ownership	Auto/manual classification: Provider/Practice, Payer, Revantage Team, No Delay/Closed.	Provides management visibility and accountability.
Billing Readiness	Final approved payer enrollment fields	Prevents approved enrollments from

	needed by billing: effective date, provider ID, group ID, EFT/ERA status, approval letter.	being missed by billing operations.
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3. High-Level End-to-End Flow



1. User logs in and lands on Dashboard.
2. Dashboard shows daily work: overdue follow-ups, missing documents, payer pending items, provider pending items, expiring documents and escalations.
3. User creates or opens provider profile and validates demographics, IDs, specialty, CAQH, NPI and professional details.
4. User maps provider to practice/location, including group NPI, TIN, billing address, service location and provider start date.
5. User collects required documents and verifies expiry/version/document type.
6. System checks whether required data/documents are complete enough to create/submit payer application case.
7. User creates payer application case and tracks status, assigned owner, aging, last action and next follow-up date.
8. User may generate an application packet when a form/fax/email based submission is required.
9. User submits application and system begins payer follow-up clock.
10. Email Center, task engine, SLA rules and delay logic support the case until approval/closure.
11. After approval, user completes billing readiness fields and notifies billing team.
12. Manager reviews reports for delays, aging, productivity and approval turnaround.

4. Roles and Permissions Flow

Role	Main usage flow	Allowed actions
System Administrator	Configures the platform before and during operations.	Manage users, roles, payers, statuses, document types, templates, SLA rules, delay rules, business calendars and system settings.
Credentialing Manager	Monitors overall credentialing operations and exceptions.	View all cases, assign/reassign work, override delay ownership with reason, approve escalations, review dashboards and reports.
Credentialing Executive	Handles daily credentialing work.	Create provider profiles, update payer cases, request documents, send emails, upload/verify documents, create/complete tasks, log follow-ups and update statuses.
Billing / Operations Read Only	Checks enrollment readiness for billing.	View approved payer status, effective dates, provider IDs, group IDs, EFT/ERA status and billing notes. No operational edits unless permission is later granted.
Provider / Practice User - optional later	Uploads requested documents and responds to requests.	Secure external login, see outstanding checklist, upload documents, view high-level status. Not required for internal MVP.

5. Main Navigation and Screen Purpose

Module	Purpose	Developer expectation
Dashboard	Operational command center.	Clickable KPIs should open filtered lists, not static numbers.
Providers	Provider master record and profile view.	All provider-related cases, documents, communications, tasks and timeline should be accessible from provider profile.
Practices	Practice/group/location master data.	Support provider-practice-location mapping without duplicating provider data.
Credentialing Tracker	Main daily work screen for payer application cases.	Filter/sort by status, payer, provider, owner, delay, aging, next follow-up and priority.
Documents	Central repository and document checklist.	Support version history, expiry dates, verification status, linking to provider/case and secure preview/download.
Email Center	Operational credentialing inbox.	Read/send/link emails, save attachments, use templates, create tasks, track unresolved threads. Not an email

		marketing analytics page.
Tasks & Follow-ups	Daily queue and task board.	Support manual and auto-generated tasks, assignments, due dates, priorities and escalation.
Reports	Manager review and performance visibility.	Aging, delay owner, productivity, approvals, document compliance, email operations and exports.
Admin Settings	Configuration.	Statuses, payers, document types, templates, SLA/delay rules, users/roles and calendars should be configurable.

6. Provider Profile Flow

6.1 Provider opening behavior

- When a user clicks a provider from the provider list, open a Provider Profile page or drawer with a persistent summary card and tabs.
- The provider profile should not be a small static card. It should be a workbench where the team can see the provider, payer applications, documents, emails, tasks, delay ownership, timeline and billing readiness.
- The top summary should show provider name, specialty, NPI, CAQH, primary practice, overall status, current delay owner, assigned executive, critical alerts and next follow-up date.

6.2 Provider profile tabs

Tab	Visible content	Main user actions
Overview	Basic demographics, provider IDs, specialty, CAQH, NPI, taxonomy, contact information, summary KPIs and alerts.	Edit profile, view missing items, jump to application/document/task.
Practice & IDs	Practice legal name, DBA, TIN/EIN, group NPI, locations, billing/pay-to addresses, provider start date, role, reassignment/relationship details.	Add practice mapping, edit location, validate group linkage.
Licenses	Medical license, DEA, CDS, board certification, malpractice, CAQH attestation, OIG/SAM/NPDB checks where applicable.	Add/verify license, set expiry, trigger reminder.
Documents	Document checklist, status, expiry, version, uploaded by, linked payer/case, preview/download.	Upload, replace, verify, reject, save email attachment, link to case.
Payer Applications	Payer cases by payer/state/type/status/owner/delay/aging/next follow-up/effective date/provider ID.	Create case, update status, add note, generate packet, submit/follow up.
Communication	Emails, call logs, faxes, internal notes and escalation notes linked to provider/case.	Reply, use template, link thread, create task, save attachment.
Tasks	Open/completed tasks with due dates, priority, owner and linked payer case.	Create, assign, complete, reopen, escalate.
Delay Tracking	Current delay owner, reason, start date, aging, SLA breach, delay history and overrides.	Manager override with reason, review delay history.
Timeline	Chronological audit of actions, documents, emails, status changes, tasks, delay changes and approvals.	Review history; should be system-generated, not manually editable.
Billing Readiness	Approved payer details: effective date, provider ID, group ID, EFT/ERA, approval letter, billing notified, notes.	Notify billing team, mark ready to bill.

7. Practice and Location Mapping Flow

- Practice/location details should be master data and reusable across providers.
- A provider can be linked to one or more practices and one or more service locations.

- A payer application should point to the specific provider-practice-location relationship being enrolled.
- Do not copy the same practice fields repeatedly into every provider case. Store master records and reference them.

Required information	Reason
Practice legal name, DBA, TIN/EIN, group NPI	Required for group enrollment and payer setup.
Billing address, service location, mailing address, pay-to address	Payer applications often require multiple address types.
Practice contact name, email, phone	Used for document requests, follow-ups and payer correspondence.
Provider start date and role	Needed for enrollment timelines and billing readiness.
Primary vs additional locations	Supports location-wise enrollment and reporting.

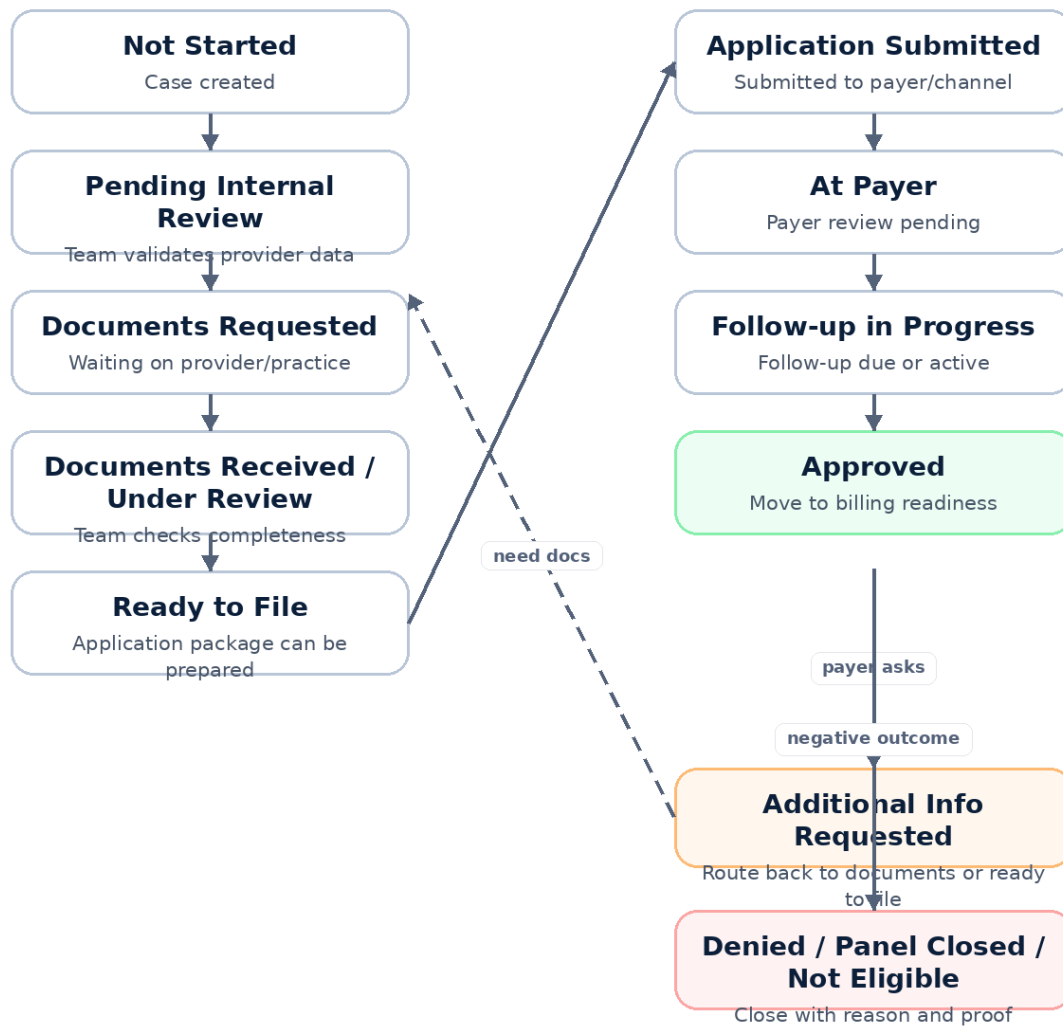
8. Document Collection and Verification Flow

- Documents may enter the system through manual upload or email attachment ingestion.
- Every document should have a document type, status, uploaded date, uploaded by, expiry date if applicable, version number and linked provider/case.
- Superseded documents should not be deleted from history. Keep version history for audit.
- Document checklist should drive readiness: if a required item is missing, the payer case should not silently proceed without warning.

Document status	Meaning	Expected system behavior
Missing	Required document is not present.	Show alert; allow document request email and task creation.
Uploaded	File exists but has not been verified.	Show pending review; do not treat as final until verified if verification is required.
Verified	Team reviewed and accepted document.	Can be used in packet/submission.
Rejected	Document is incorrect, unclear or expired.	Require rejection reason and create follow-up if needed.
Expired	Expiry date has passed.	Create alert and renewal task.
Superseded	Older version replaced by newer version.	Keep in version history and audit trail.

9. Credentialing Case Flow

Credentialing Case Status Flow



9.1 Case creation

- A payer application case is created when the team is ready to track a specific enrollment/credentialing activity.
- A provider may have many cases: multiple payers, multiple states, multiple practices/locations, and multiple application types.
- The system must prevent duplicate open cases for the same provider + practice/location + payer + state + application type unless the user confirms an exception.

Case field	Purpose
Case ID	Unique internal identifier for linking emails, tasks, documents and timeline.
Provider	Provider being credentialed/enrolled.

Practice and location	Group/location relationship for the case.
Payer and state	Payer-specific workflow tracking.
Application type	Credentialing, recredentialing, reassignment, EFT, ERA, roster, group enrollment, maintenance etc.
Status	Configurable workflow state.
Assigned executive / manager	Operational ownership.
Intake date, submission date, acknowledgement date, expected completion date	Aging and SLA calculation.
Last action date, next follow-up date	Daily work queue control.
Delay owner and delay reason	Accountability and reporting.
Effective date, provider ID, group ID, approval letter	Billing readiness.

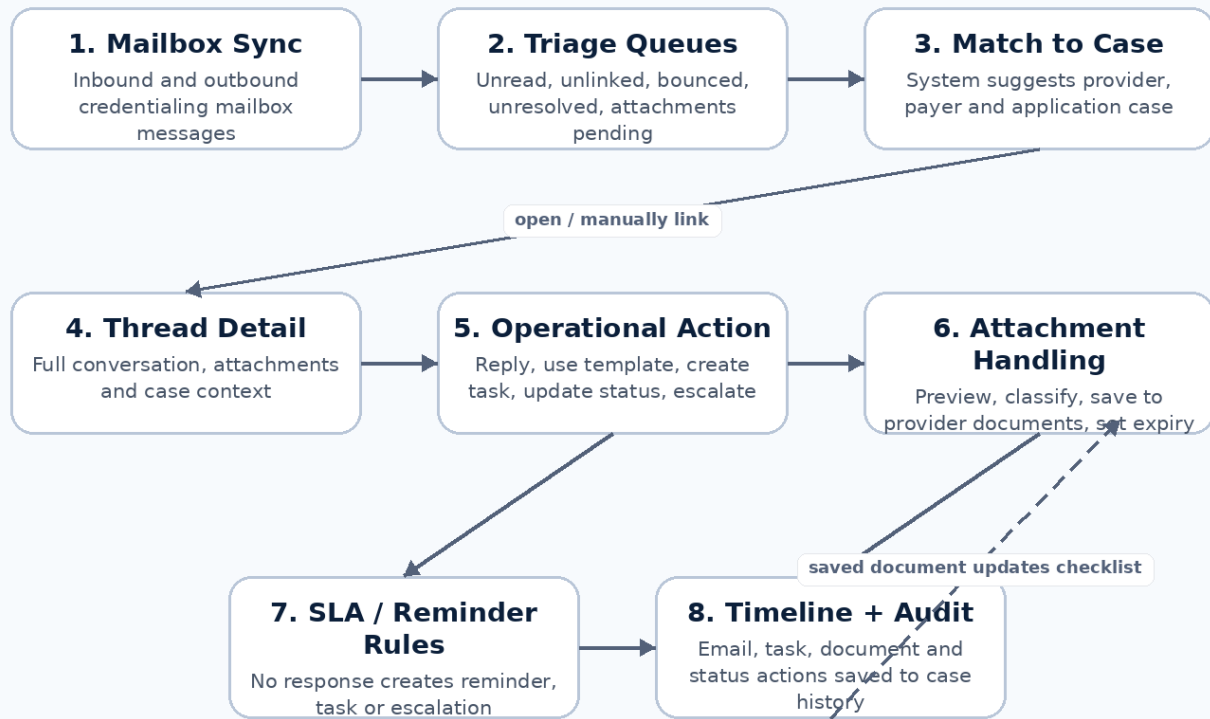
10. Credentialing Tracker Flow

- The Credentialing Tracker is the main daily work screen, similar to an operations queue.
- It should not be only a listing page. It should allow fast updates, filtering, bulk review, and drill-down into application detail.
- Each row represents one payer application case, not only one provider.

Tracker capability	Developer notes
Filters	Pending with provider, pending with payer, overdue, recently submitted, escalated, approved, on hold, revalidation due, owner, payer, state.
Inline fields	Status, assigned to, priority, next follow-up date, delay owner, aging, last action.
Detail drawer	Open case detail without losing tracker filter context. Show documents, emails, tasks, notes, timeline and status history.
Bulk actions	Optional later: assign owner, export filtered list, set follow-up date, mark selected cases for review.
Audit logging	Every status/owner/date update should create a timeline entry.

11. Email Center Flow - Correct Interpretation

Email Center Operational Flow



Not an analytics page: this screen must help the team work emails, link conversations to cases, save attachments and trigger follow-ups.

Important correction

Email Center should not be designed as an email analytics or marketing engagement page. It should be an operational credentialing inbox connected to provider and payer application cases.

11.1 What the Email Center must do

- Sync credentialing mailbox inbox and sent items.
- Show unread, unresolved, unlinked, bounced/failed, provider responses, payer responses and attachment-pending queues.
- Allow user to open a full thread and see the linked provider, linked payer application, current case status, last action and next follow-up date.

- Allow manual link when auto-matching is uncertain.
- Allow user to save attachments directly into the provider document repository with document type, expiry date, status and linked payer/case.
- Allow reply/forward/compose from the credentialing mailbox using approved templates.
- Allow create task from email and update case status from email context.
- Log email actions in provider/case communication history and audit timeline.

Email Center queue	Purpose
Inbox	New and active credentialing emails.
Unlinked Emails	Emails system could not confidently match to provider/case. Must be triaged daily.
Provider Responses	Replies from providers/practices, often with documents or information.
Payer Responses	Payer replies, reference numbers, approvals, additional information requests.
Attachments Pending Review	Emails with attachments not yet classified/saved to documents.
Replies Awaited	Outbound document requests or payer follow-ups with no response.
Escalation Queue	Threads that crossed configured reminder/escalation rules.
Failed/Bounced	Emails that failed delivery or need corrected recipient.
Templates	Document request, reminder, payer follow-up, escalation, approval notice, billing notification, revalidation reminder.

11.2 Email-to-case matching rules

Matching signal	How system should use it
Case ID in subject/body	Highest confidence direct link to case.
Provider name / NPI / CAQH	Match to provider. If multiple cases exist, ask user to pick payer/case.
Payer name / payer domain / known payer contact	Suggest payer case match.
Sender email	Match practice/provider/payer contacts from master data.
Attachment filename	Suggest document type and provider if file name contains provider/document clues.
Thread history	Maintain link once thread is linked; new replies should inherit the case link unless user changes it.

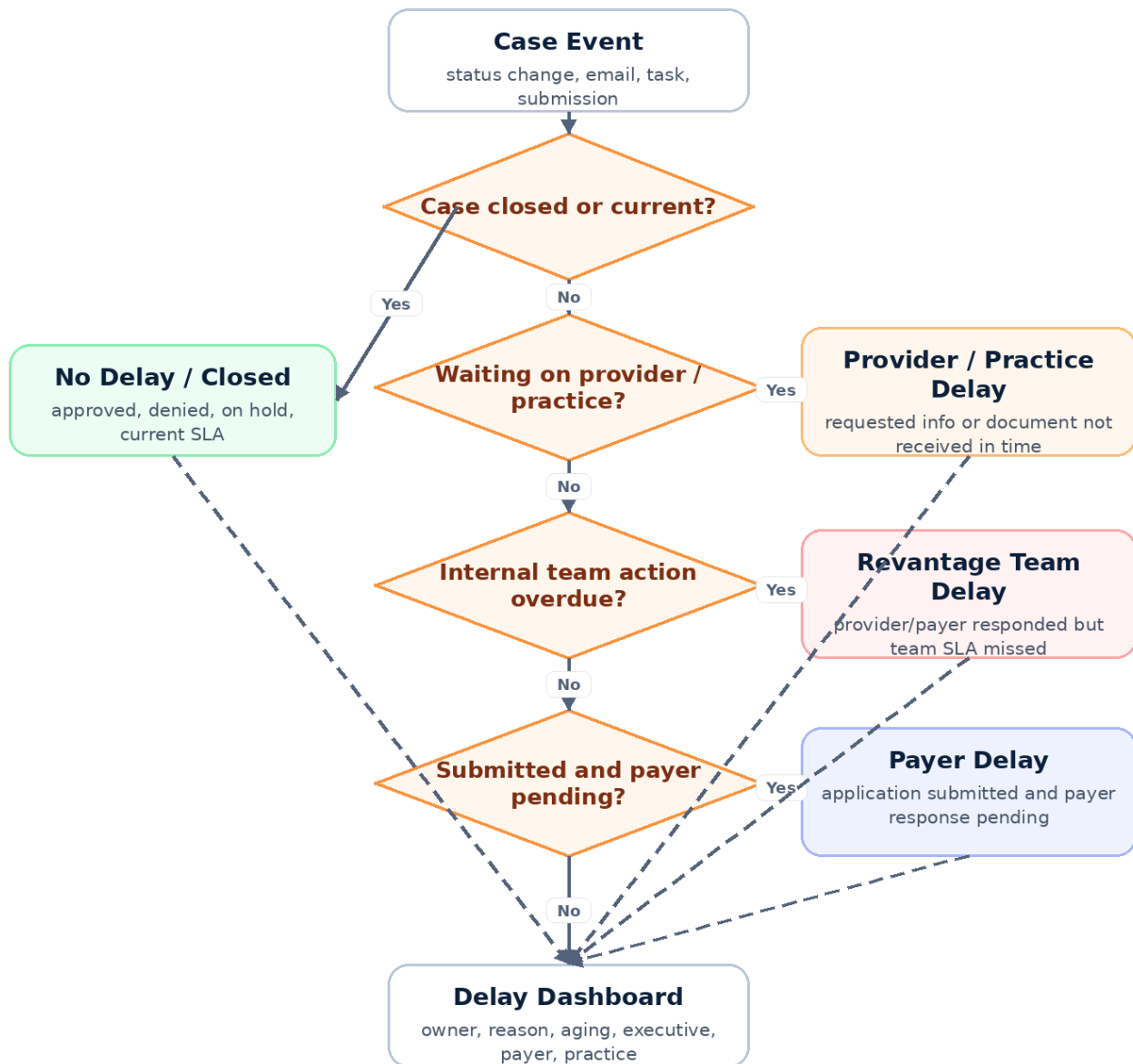
12. Task and Follow-Up Flow

- Tasks should be created manually by users and automatically by system rules.
- Each task must link to provider and, where possible, a payer application case.
- Task board should show Due Today, Upcoming, Overdue, Completed and Escalated.
- Completing a task should create a timeline entry and may update last action date on the case.

Trigger	System action
Document request sent	Create follow-up task due after configured business days.
Provider reply received	Create review task or mark existing document request task as ready for review.
Application submitted	Create payer follow-up task based on payer/SLA rule.
No payer response	Create next follow-up task or escalation task.
License/document expiring	Create renewal task and notification.
Approval received	Create billing readiness task if effective date/provider ID/EFT/ERA/approval letter not complete.
Revalidation approaching	Create maintenance/revalidation task.

13. Delay Ownership Flow

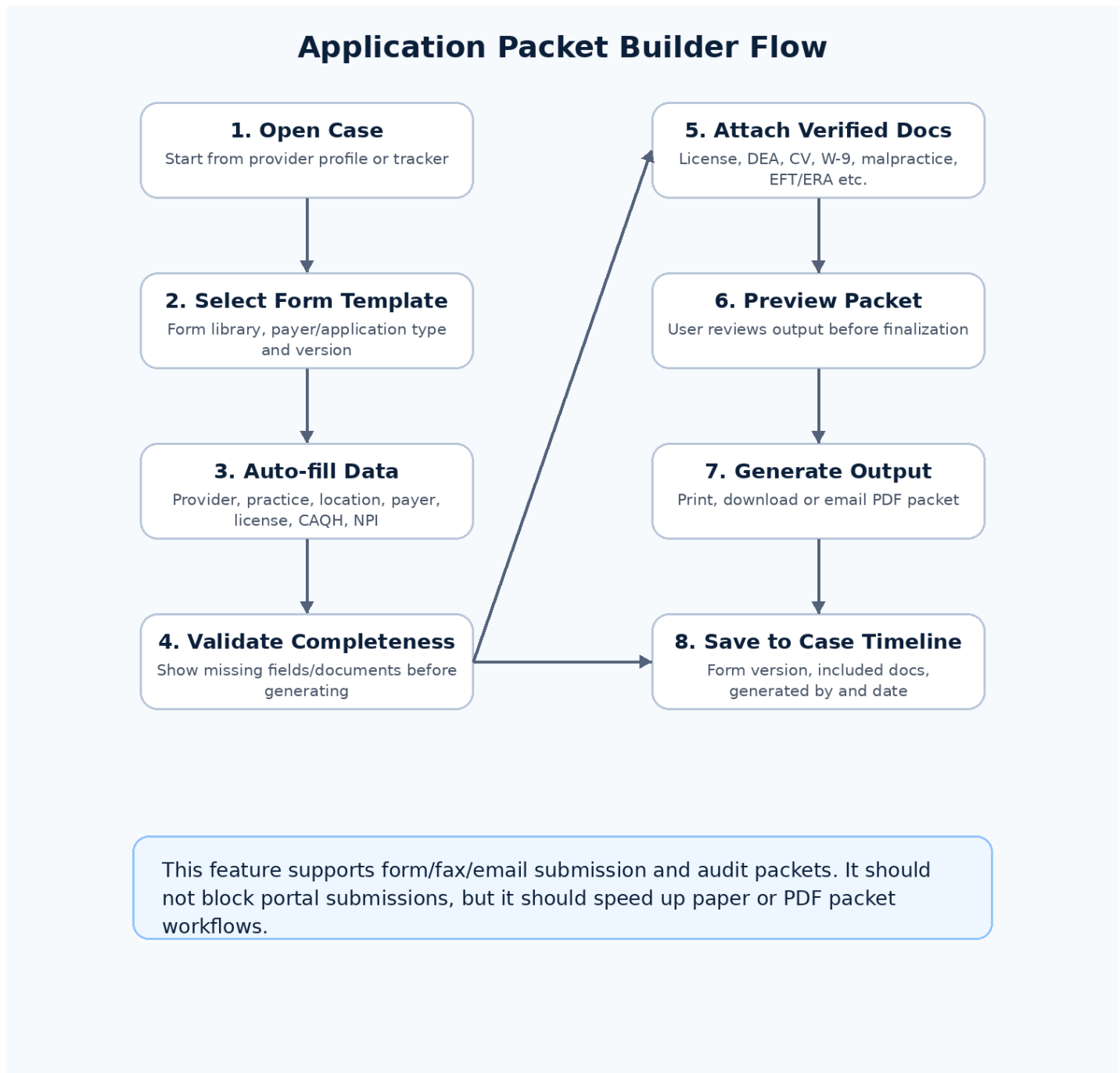
Delay Ownership Decision Flow



Delay owner	When applied	Example
Provider / Practice	Team requested provider/practice information or document and no response is received within configured SLA.	DEA copy requested; no response after 3 business days.
Revantage Team	Provider/payer has responded but internal action was not completed within defined SLA.	Provider sent documents, but team has not reviewed them within 2 business days.
Payer	Application submitted and waiting on payer response, and internal follow-up SLA has not been missed.	Application at payer; no payer decision yet.
No Delay / Closed	Case is current, approved, denied, closed, on valid hold, or not yet breaching	Application submitted yesterday; next follow-up not due.

- Delay owner should be calculated as first-class business logic, not only a report filter.
- Managers may override delay owner for exceptions, but override reason and user must be logged.
- Delay dashboard should show aging by delay owner, executive, payer, provider, practice and state.
- Business-day calendar should be used for SLA calculation. Weekends/holidays should be configurable.

14. Application Packet Builder Flow



- This feature supports the competitor-style workflow: select form, select provider/case, generate populated credentialing packet.

- It should not replace online submissions but should support PDF/fax/email/form-based workflows and internal review packets.
- Generated packets must be saved to the case timeline with form version, included documents, generated by and generated date.

Step	Developer behavior
Select case	User starts from provider profile or credentialing tracker and selects the relevant payer application case.
Select form/template	User chooses approved form/template from Admin-managed library.
Auto-fill fields	System pulls provider, practice, location, payer, license, CAQH, NPI, taxonomy and case data.
Validate completeness	System highlights missing fields/documents before generating packet.
Attach supporting docs	System suggests required verified documents based on payer/form/application type.
Preview	User reviews before output.
Generate PDF	System generates print/download/email packet.
Save history	Save generated PDF and packet metadata to provider/case timeline.

15. Approval and Billing Readiness Flow

- Approval is not the end of the workflow. The case must be converted into billing-ready payer enrollment information.
- The system should require key fields before marking a payer case as Ready to Bill.
- Billing/Operations read-only users should be able to see final enrollment details without editing credentialing history.

Billing readiness field	Purpose
Approval date	When approval was received.
Effective date	Date billing can start from. Critical for claims.
Provider ID	Payer assigned rendering/provider identifier.
Group ID	Payer assigned group identifier where applicable.
EFT status	Confirms payment enrollment.
ERA status	Confirms electronic remittance enrollment.
Approval letter uploaded	Proof of payer approval.
Billing team notified	Confirms handoff from credentialing to billing.
Billing notes	Any payer-specific billing limitations or instructions.

16. Reporting and Manager Review Flow

Report / dashboard	What it answers
Applications by status	How many cases are in each status and where the pipeline is stuck.
Aging report	Which cases have been open/submitted/waiting longest.
Delay ownership report	Whether delays are with provider/practice, payer or internal team.
Executive productivity	Follow-ups completed, cases handled, approvals achieved, overdue tasks.
Document compliance	Missing and expiring provider documents.
Email operations	Unlinked emails, pending replies, bounced emails, reminders sent, unresolved threads.
Approval/denial trends	Payer and provider trends for management review.
Revalidation upcoming	Future maintenance work due in next 30/60/90 days.

17. Admin Configuration Flow

- Admin Settings is critical because credentialing requirements change frequently.
- Avoid hard-coding statuses, document types, SLA days, payer rules and email templates into the application code.
- Admin changes should be logged and, where important, versioned.

Admin area	Configurable items
Users and roles	Permissions, role-based menus, read/write access, manager/admin controls.
Payers	Payer name, state, contact info, portal details, submission channel, turnaround expectation.
Statuses	Status labels, active/inactive, order, dashboard category, delay mapping.
Document types	Required flags, expiry rules, accepted file types, checklist mapping.
SLA rules	Business-day rules, provider reminder days, payer follow-up days, escalation days.
Delay rules	Mapping from statuses/events to delay owner and aging clock.
Email templates	Document request, reminders, payer follow-up, escalation, approval, billing notification.
Form templates	Application packet forms, field mapping, required supporting documents, version control.
Notification rules	Who receives alerts for overdue tasks, expiring docs, escalations, failed emails.
Business calendar	Holidays/weekends for SLA calculation.

18. Event and Audit Logging Requirements

Event	Should log
Provider created/updated	User, date/time, fields changed, old/new values where practical.
Practice/location mapping changed	Relationship added/removed/edited.
Document uploaded/replaced/verified/rejected/downloaded	Document type, version, user, date/time, linked case, rejection reason if any.
Case status changed	Old status, new status, user, date/time, comment.
Task created/assigned/completed/reopened	Owner, due date, status, linked case.
Email sent/received/linked/unlinked	Thread ID, sender/recipient, subject, linked provider/case.
Delay owner changed	Old owner, new owner, rule/manual override, reason.
Packet generated	Template, form version, included docs, generated by/date.
Approval/billing readiness updated	Effective date, provider ID, group ID, billing notified status.
Admin configuration changed	Config item, old value, new value, admin user.

19. Recommended Backend / API Flow

- Use API-driven architecture with normalized SQL Server schema.
- Keep master configuration separate from transaction records.
- Use event-driven service logic for activity log, notifications, delay recalculation and follow-up creation.
- Do not let the UI calculate business-critical aging/delay logic independently. Backend should be source of truth for SLA and delay ownership.
- For email integration, use approved Microsoft Graph/Gmail Workspace API or another approved enterprise-safe method. Avoid storing mailbox passwords insecurely.

Service / API area	Expected responsibility
Authentication/RBAC	Login, sessions, password policy, optional MFA, menu/action permissions.
Provider service	Provider CRUD, IDs, specialties, CAQH, NPI, search.
Practice/location service	Practice master, location master, provider-practice-location mapping.

Credentialing case service	Case CRUD, status, assignment, aging, next follow-up, priority, outcome.
Document service	Upload, secure storage, versioning, preview/download, expiry metadata, antivirus/file validation.
Task service	Manual tasks, auto tasks, assignment, due dates, task board filters.
Email service	Sync inbound/outbound, send/reply/templates, thread linking, attachment ingestion, failed/bounced handling.
Automation/SLA service	Reminder creation/sending, escalation, delay owner calculation, business-day calendar.
Packet builder service	Template mapping, auto-fill, validation, PDF generation, save generated packet.
Reporting service	Dashboards, exports, productivity, aging, delay, document and email reports.
Audit service	Immutable/material event history and timeline generation.

20. Key Edge Cases Developer Should Handle

- Same provider has multiple payer cases open at the same time. Emails/documents must link to the correct case.
- Same provider works under multiple practices or locations. Payer case must reference the correct practice/location relationship.
- Provider sends multiple documents in one email. User must classify each attachment separately.
- Payer sends approval but effective date or provider ID is missing. Case should not become billing ready until required fields are complete.
- An automated reminder should not send if the case is approved, denied, terminated, manually on hold, or marked do-not-automate.
- A thread may be unlinked at first and linked later. Once linked, prior messages should appear in the case history.
- A document is replaced after submission. The system should preserve what version was used in the original packet/submission.
- Manager override of delay owner should require reason and audit log.
- Duplicate provider records should be detected by NPI and name/DOB combination. Merge process can be post-MVP but duplicate warning is important.
- SLA calculation should respect business days and configured holidays.

21. MVP Acceptance Criteria for Developer

Area	Acceptance criteria
Provider setup	User can create provider, add IDs, link practice/location and see provider profile tabs.
Credentialing case	User can create payer application case and track status, owner, dates, aging, next follow-up and delay owner.
Documents	User can upload, preview/download, classify, verify/reject, version and track expiry.
Tasks	User can create manual tasks; system can generate basic follow-up/expiry tasks.
Email Center	User can view synced emails, send/reply from portal, link thread to provider/case, save attachments, use templates and create task from email.
Delay ownership	System calculates delay owner using configured rules and manager can override with reason.
Timeline	Material actions are logged and visible in provider/case timeline.
Billing readiness	Approved case requires effective date, provider ID/group ID where applicable, approval letter and billing notification status.
Reports	Manager can view aging, delay, status and productivity dashboards with export capability.
Admin settings	Statuses, document types, payers, templates and SLA rules are configurable without code changes.

22. Open Decisions to Confirm with Credentialing Team

- Exact provider document checklist by specialty and payer type.
- Default SLA days for provider reminders, payer follow-ups, internal review and manager escalation.
- Which mailbox integration will be approved: Microsoft 365/Graph, Gmail Workspace API or other approved method.
- Whether automated emails should send automatically or only create a suggested email/task for user review in MVP.
- Required fields for Ready to Bill for each payer type.
- Whether external provider/practice portal should be included in MVP or remain post-MVP.
- Which forms/templates should be included first in Application Packet Builder.
- Whether packet builder output must support fillable PDFs, flattened PDFs, or both.
- Retention rules for documents, email logs and audit records.
- Named roles/team members for assignment logic and escalation notifications.

23. Developer Build Sequence Recommendation

13. Build authentication, RBAC, app shell and role-based navigation.
14. Build master data: providers, practices, locations, payers, statuses, document types.
15. Build provider profile and practice/location mapping.
16. Build credentialing case/tracker with status, owner, dates, aging and timeline logging.
17. Build document upload/versioning/expiry/checklist.
18. Build task board and work queue.
19. Build email center as operational inbox and case-linking workflow.
20. Build delay ownership and SLA engine.
21. Build billing readiness handoff.
22. Build reports and exports.
23. Build packet builder/form template library.
24. Harden security, audit logging, backup/restore and admin configuration.